

LN2 Quiz: National accounts and growth

1. GDP accounting

Which expression is the national income identity?

- (A) $Y - IM = C + I + G$
- (B) $Y + EX = C + I + G + IM$
- (C) $Y - EX = C + I + G$
- (D) $Y + IM = C + I + G + EX$

2. Nominal and real GDP

Nominal GDP and the GDP deflator both increase by 10 percent. What happens to real GDP?

- (A) It increases by approximately 20 percent.
- (B) It increases by 10 percent.
- (C) It remains unchanged.
- (D) It decreases by 10 percent.

3. Constant returns to scale

Which **two** of the following production functions display constant returns to scale?

- (A) $Y = F(K, L) = K^{\frac{1}{2}} + L^{\frac{1}{2}}$
- (B) $y = f(k) = k$
- (C) $Y = F(K, L) = KL$
- (D) $Y = F(K, L) = K^{\frac{1}{2}} L^{\frac{1}{2}}$
- (E) $Y = F(K, H, L) = K^{\frac{1}{2}} H^{\frac{1}{2}} L^{\frac{1}{2}}$

4. Growth accounting

Output per capita grows by 2 percent and capital per capita grows by 3 percent. If the capital share is $1/3$, approximately how fast does TFP grow?

- (A) 0 percent
- (B) 1 percent
- (C) 2 percent
- (D) 3 percent

5. Solow model

Which equation best describes how the capital stock develops over time in the Solow model?

- (A) $K_{t+1} = K_t + sAK_t^\alpha L_t^{1-\alpha} - \delta K_t$
- (B) $K_{t+1} = (1 - \delta)K_t + Y_t + I_t$
- (C) $K_{t+1} = (1 - \delta)Y_t + I_t$
- (D) $K_{t+1} = (1 - \delta)K_t + sI_t$

6. A poverty trap

At which level of capital will the economy end up if initial capital is K_3 ? If initial capital is K_5 ?

- (A) $K_3 \rightarrow K_4$ and $K_5 \rightarrow K_6$
- (B) $K_3 \rightarrow K_2$ and $K_5 \rightarrow K_4$
- (C) $K_3 \rightarrow K_4$ and $K_5 \rightarrow K_4$
- (D) $K_3 \rightarrow K_2$ and $K_5 \rightarrow K_6$

